



PROTEINS

*These are organic compounds of large molecular mass and insoluble in water. In addition to **C,H** and **O**, they always contain **N**, usually **S** and sometimes **P**.*

- ❖ Whereas there are few carbohydrates and fats, the *number of proteins is limitless* e.g. a single bacterium may have around **800 types** of proteins while man has **10,000 types**.
- ❖ This is because there are several amino acids which may join in different patterns hence forming the various types of proteins.

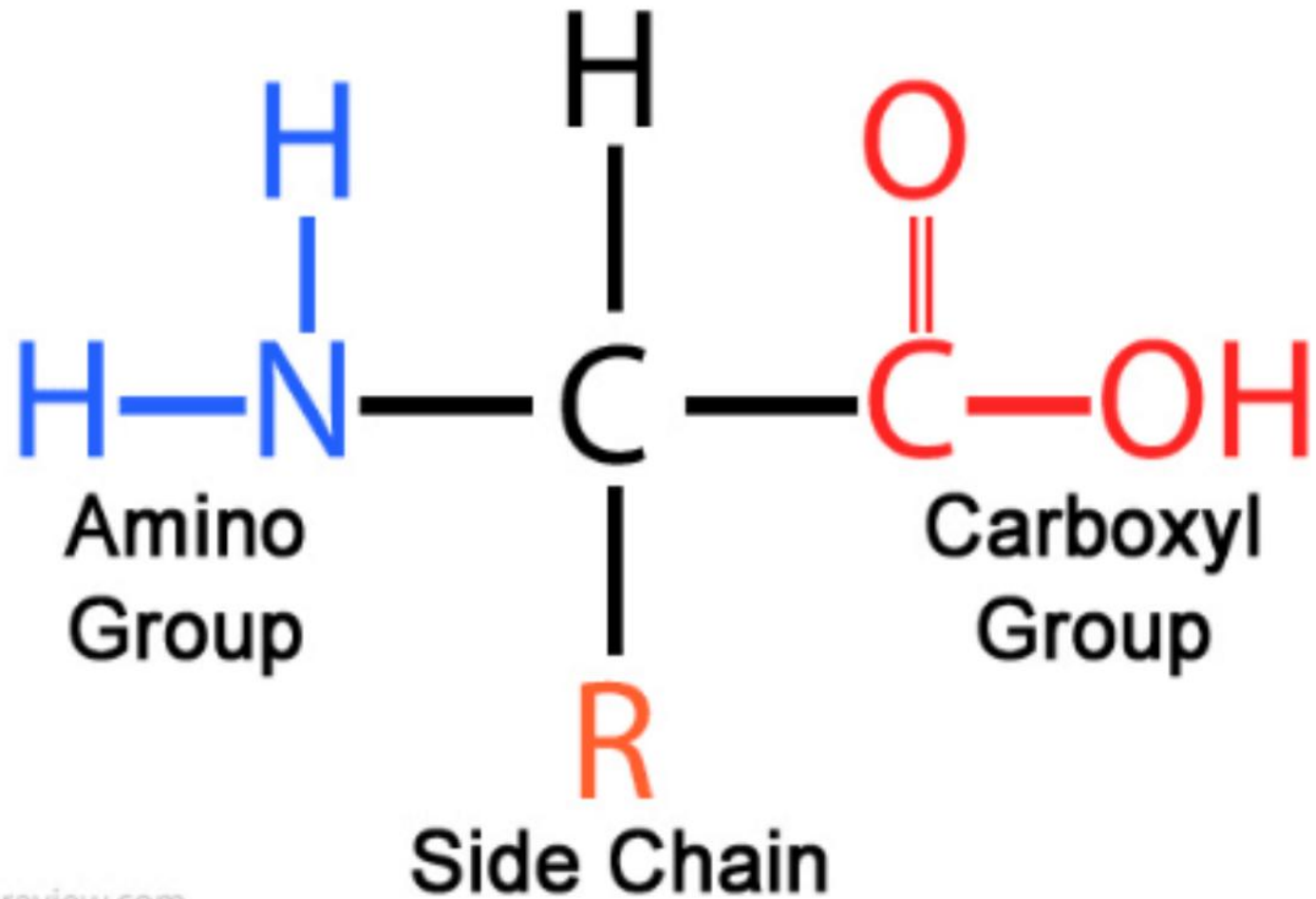
PROTEINS

- ❖ Proteins are **specific** to each species hence determine the character of the species.
- ❖ Proteins are **not stored** in the organism except in **eggs** and **seeds** where they are used to form new tissues.
- ❖ Proteins form the **structural** basis of all **living cells**.
- ❖ Their building blocks are the **amino acids**.

AMINO ACIDS

- ✓ These are groups of many chemicals of which **around 20** occur in proteins. They contain an amino group (**NH₂**) and a carboxyl group (**COOH**).
- ✓ Most **amino acids** have one of each and are therefore **neutral** but a few have more amino groups than carboxyl making them alkaline or may have more carboxyl than amino groups making them acidic.

Amino Acid Structure

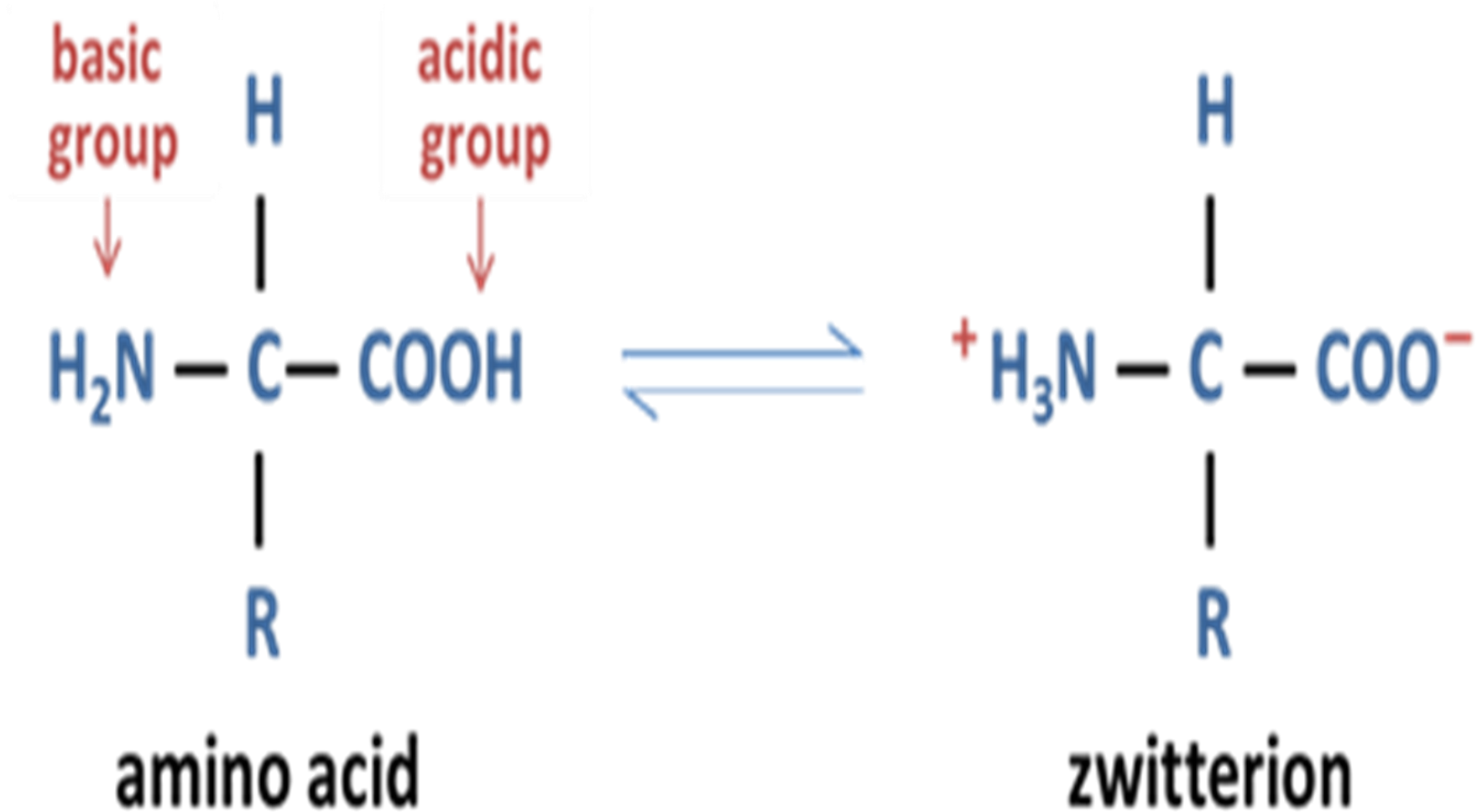


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Amino acids

- Amino acids are **soluble** in water and ionize to form ions.
- The carboxyl end of the amino acid is acidic in nature. It will ionize in water to give H^+ . This will make the **COOH group** negatively charged.
- The amino end (**NH_2**) is basic in nature. It attracts the H^+ in solution making it positively charged. The ion is now **dipolar** i.e. having a **negative** and a **positive pole**. Such ions are called **zwitter ions** i.e. the **negative** and **positive** charges exactly **balance** and the amino acid ion has no overall charge.

Zwitter ion (no overall charge)



Amino acids

- ❖ Therefore in acidic solutions, an amino acid acts like a base and in alkaline solutions, it acts as an acid. In neutral conditions found in the cytoplasm of most living organisms, the amino acid acts as both.
- ❖ Amino acids therefore show both **acidic** and **basic properties** i.e. they are **amphoteric**.
- ❖ The overall charge of the amino acid depends on the pH of the solution.
- ❖ At some characteristic pH, the amino acid has no overall electric charge i.e. it exists as a zwitter ion. This pH is called the **isoelectric point** of an amino acid.

Amino acids

- ❖ If the pH falls **below the iso-electric point** i.e. the solution becomes **more acidic**, H^+ are taken up by the carboxyl ion. This reduces the concentration of the H^+ in solution making the solution less acidic and the amino acid gains an overall positive charge.
- ❖ If the pH rises **above the iso-electric point** i.e. it becomes less acidic or more **alkaline**, hydrogen ions are lost by the amino group. This increases the concentration of free H^+ in the solution making it more acidic and the amino acid gains an overall negative charge. Therefore being **amphoteric**, amino acids are buffers.

NOTE:

a buffer solution is one which resists the tendency to alter its pH even when small amounts of acid or base are added to it.

Question

how do amino acids act as buffer solutions?

TYPES OF AMINO ACIDS

1. Essential Amino acids

These are amino acids that cannot be synthesized by the body and therefore got from the diet that the organism feeds on. They include:

- ❖ **Histidine**
- ❖ **Isoleucine**
- ❖ **Leucine**
- ❖ **Proline**
- ❖ **Phenylalanine**
- ❖ **Valine**
- ❖ **Arginine**
- ❖ **lysine**
- ❖ **Methionine**
- ❖ **Tryptophan**

TYPES OF AMINO ACIDS

2.Non-Essential amino acids

These are amino acids that are synthesized by the body through a process called transamination

They include:

- ❖ **Tyrosine**
- ❖ **Alanine**
- ❖ **Glycine**
- ❖ **Serine**
- ❖ **Theonine**
- ❖ **Cystine**
- ❖ **Cystein**
- ❖ **Aspartic acid**
- ❖ **Glutamic acid**
- ❖ **Asparagine**

Proteins can be classified into:

❑ **First class proteins**

which contain **all** the **essential** amino acids e.g. from beef.

❑ **Second class proteins**

which are *deficient* of one or more essential amino acid.

FORMATION OF POLYPEPTIDES

They are formed as a result of **condensation reaction** between the amino group of one amino acid and the ***carboxyl group*** of another amino acid to form ***a dipeptide***.

Further combinations of this type extend the length of the chain to form ***a polypeptide*** which usually contains many amino acids.

NB:

The shape of the polypeptide molecule is due to four types of bonding which occur between the various amino acids in the chain. These bonds include:

Disulphide bond:

It is formed between sulphur containing groups on any two cysteine molecules.

Ionic bond:

It is formed between **NH₃⁺** and **COO⁻** groups.

Hydrophobic interactions:

These are interactions between non polar ***R groups*** which cause the proteins to fold as hydrophobic side groups.

The polypeptide chain showing 3 types of bonding

